

Contact Forces: Muscular Force and Friction

Forces that act only when objects touch — the muscular force we use and the friction that opposes motion.

Forces that need touch

Forces that act only when there is **physical contact** between objects are **contact forces**. (Contact can be direct — a hand — or indirect — a stick or rope.)

Muscular force

The force from the action of our **muscles** (walking, lifting, pushing) is **muscular force**. It works as muscles contract and relax. It also runs body processes — chewing, pushing food along the alimentary canal, and the heart pumping blood. Animals also use muscular force.

Friction

A rolling ball stops on its own because of **friction**. **Friction is the force that acts when an object moves, or tries to move, over another surface.** It always acts **opposite** to the direction of motion. It is a **contact force**.

- **Why it happens:** every surface has tiny **irregularities**; they lock into each other and oppose sliding.
- **What it depends on:** the **nature of the surfaces** — friction is **greater on rough surfaces**, less on smooth ones.

Friction in fluids: air and water also exert friction. That's why aeroplanes, ships and fast trains have streamlined shapes to reduce it.

Where students lose marks

Trap 1 — Friction opposes motion. It always acts opposite to the direction the object moves or tries to move — never in the same direction.

Trap 2 — Friction is a contact force. It needs two surfaces touching; it isn't a "non-contact" force.

Trap 3 — Rough = more friction. A rougher surface gives more friction; a smoother one gives less.

Even fluids: air and water cause friction, so vehicles are streamlined.

Model answers — write them like this

Example A. Why does a ball rolling on the ground slow down and stop?

Step 1: The surfaces of the ball and ground are in contact.

Step 2: Friction acts opposite to motion, reducing the speed until it stops.

∴ Friction opposes its motion and brings it to rest.

Example B. Why does the same box slide a shorter distance on sand than on glass?

Step 1: Sand is rougher than glass.

Step 2: Rougher surfaces give more friction.

∴ More friction on sand stops the box sooner.

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